Walnut Technologies

PRODUCT CATALOG 2026

Intelligent Embedded Systems, Engineered for a Connected World

Medical · Neuro-rehabilitation · Fintech · Automotive · IoT



Walnut Technologies Pvt. Ltd. · Mohali, India · walnutmedical.in

CONTENTS

Inside this catalog

Walnut Medical & Walnut Technologies design and manufacture electronics end to end — from idea and PCB design to SMT, assembly, testing and dispatch. This catalog introduces the company and every product line we build.

01 **About Walnut** — Who we are, at a glance, leadership — 03

02 **Our journey** — 2016 – 2026 milestones — 06

03 **Infrastructure & quality** — Facilities, production process, certifications — 07

04 **Medical devices** — BP monitor, thermometer, nebulizer, oxygen concentrator — 10

05 **Neuro-rehabilitation** — WalkLab, Weebo, CES Repose — 14

06 **Fintech & payment devices** — Soundbox platform and models — 17

07 **Automotive electronics** — 5″ & 7″ clusters, Google Maps mirroring, tractor suite — 20

08 **IoT solutions** — GPS tracker & gateway, 4G smart lock — 25

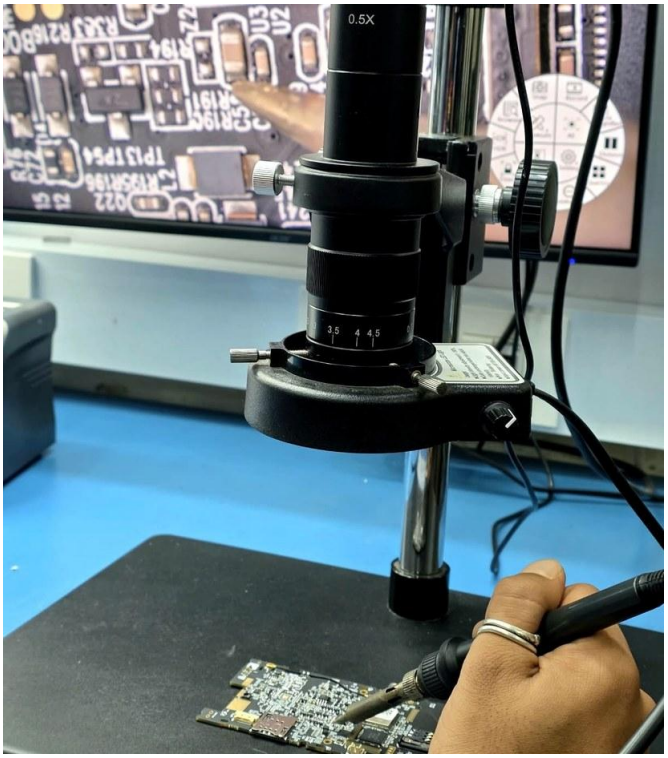
09 **Expertise & clients** — Capabilities, brands and industries — 28

10 **Contact** — Addresses and contacts — 30

Who we are

Walnut Medical & Walnut Technologies is a multi-vertical original design and equipment manufacturer (ODM/OEM). Established in 2016, we have grown from a small electronics workshop into a vertically integrated design and manufacturing partner for medical devices, neuro-rehabilitation robotics, payment devices, IoT and automotive electronics.





Production units

Mohali (Punjab) · AMTZ, Visakhapatnam (Andhra Pradesh) · Dhanana (Upcoming in Haryana)

Fully in-house

Electronics & firmware design, mould design, SMT, testing, quality and dispatch

Certified quality

ISO 13485 · Class 10K cleanroom · CE · FCC · BIS

Walnut at a glance

10+ Years of excellence	**600+** Engineers & technicians	**10+** Countries served
500K+ Units / month	**150,000** Sq. ft facility	**4** SMT lines

Mission

Deliver world-class original design manufacturing for electronics across healthcare, finance and technology — globally.

Vision

The most trusted ODM partner for mission-critical electronics, ISO 13485 certified, serving 50+ countries by 2030.

OUR VALUES

Quality first

ISO 13485 and ISO 9001 certified processes with a Class 10K cleanroom.

Innovation

600+ engineers pushing boundaries in medical, payment and custom electronics.

Global reach

Serving 10+ countries with end-to-end manufacturing and logistics.

Vertical integration

From PCB design to final assembly — complete in-house capabilities.

Leadership



SD

Siddharth Dhawan

Director & CEO

Spent 10 years at National Instruments Corp. in the USA and India, working in high-tech embedded and control systems and managing the business in India.

AB

Amol Bhardwaj

Director & CTO

Leads medical device development; launched devices across diagnostics, therapeutics and robotics — EEG, ECG, rTMS and gait training robots. Expert in QC tools, SPC, APQP, PPAP, Six Sigma, Cpk and 5S; Management Representative for ISO 13485 and CE audits (2009–2014).

Our journey

2016

Walnut Medical founded

Started with a vision to make quality healthcare more accessible.



2017

Robotics & neuro-rehab

Walkex functional therapy and the WalkLab gait-training robot.



2019

Medical devices

IR thermometer and digital BP monitor, then nebulizers and oxygen concentrators.

2021

Walnut Technologies

Engineering expands beyond healthcare into electronics for every industry.



2022

Fintech · soundbox

Smart payment soundboxes and POS devices, in mass production from 2023.



2025

Smart clusters

5″ and 7″ smart instrument clusters for 2-wheelers, tractors and vehicles.



2026

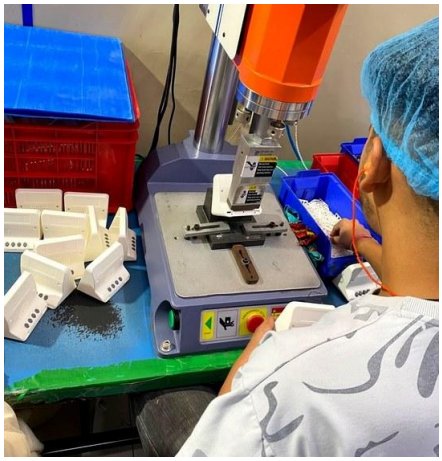
GPS tracker & IoT

GPS tracker & gateway and a growing range of connected IoT devices.

Manufacturing infrastructure








150,000 sq. ft facility

Mohali · AMTZ Visakhapatnam · Dhanana (Upcoming in Haryana)

4 SMT production lines

Fully automatic SMT — JUKI RS-1, 100,000 CPH

In-house test & calibration

Keysight · National Instruments · Rigol · Tektronix

PCB design to production

Design, validation, prototyping, pilot lots, full scale

IQC · LQC · OQC quality teams

Class 10K cleanroom for controlled builds

Moulds & injection moulding

Mass-production mould design and plastics in-house

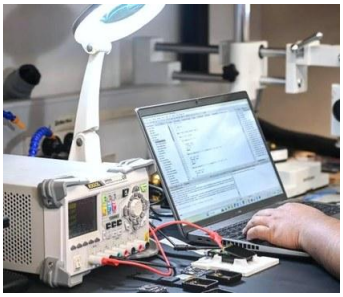
End-to-end production process



01

Idea & Requirement

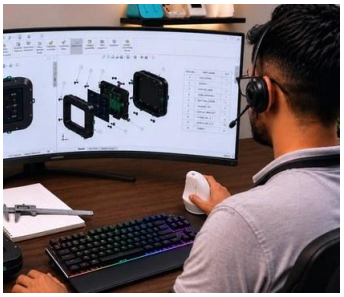
Clear direction



02

Research & Market Analysis

Right opportunity



03

Design & Engineering

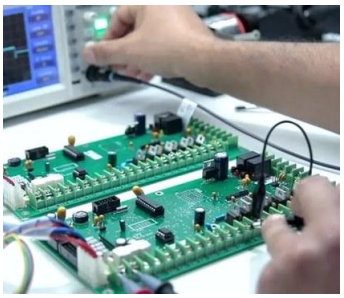
Optimized design



04

Prototype Development

Working prototype



05

Validation & Testing

Validated performance



06

Manufacturing Engineering

Production ready



07

Quality Assurance

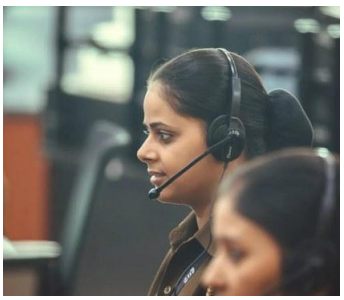
Zero-defect mindset



08

Packaging & Dispatch

Safe, on-time delivery



09

After-Sales Support

Reliable support



10

Continuous Improvement

Better every day

Quality & certifications



Quality is built into every stage — incoming, in-line and outgoing inspection, environmental and functional testing, and full traceability from component to customer.

- Incoming inspection (IQC)
- In-process inspection (LQC)
- Final quality check (OQC)
- Reliability & environmental testing
- Traceability

ISO — **ISO 13485**
Quality management for medical devices

10K — **Class 10K cleanroom**
Controlled manufacturing environment

CE — **CE · FCC · PCI-DSS**
European, US & payment-security standards

IEC — **IEC 60601**
Medical electrical equipment safety

BIS — **BIS certified**
Bureau of Indian Standards

ISO — **ISO 9001**
Quality management systems

Compliances: ISO 13485, ISO 80601-2-69, IEC 60601-1, IEC 60601-1-2 (NABL-accredited labs), CE, FCC, RoHS

Walnut Technologies

04

Medical devices

CDSCO-approved Class B and Class C medical devices, designed and manufactured under ISO 13485 in a Class 10K cleanroom.

CDSCO approved | ISO 13485 | IEC 60601 | Class 10K cleanroom









Digital Blood Pressure Monitor



Clinical-grade precision

Precision blood pressure monitoring device with advanced cuff technology and a digital display for accurate clinical readings. Memory storage for multiple readings and an easy-to-use interface. A gold-standard electronic BP monitor is in development with ICMR funding and clinical trials at AIIMS Nagpur, AIIMS Delhi and Chettinad Super Speciality Hospital.

KEY FEATURES

- Clinical grade accuracy
- Memory storage
- Systolic, diastolic & pulse
- Large digital display
- NABL certified
- Simple one-touch operation

CDSCO approved **ISO 13485**

Respiratory & diagnostics



IR Thermometer

ThermoSure · non-contact temperature measurement

Fast, hygienic temperature measurement with infrared technology. Ideal for clinical screening and home use — instant readings without physical contact.

- Non-contact
- Instant reading
- Hygienic
- Battery operated



Compressor Nebulizer

Respiratory therapy

Efficient compressor nebulizer for respiratory therapy. Compact design with reliable performance for clinical and home settings, delivering medication directly to the lungs.

- Efficient nebulization
- Low noise
- Compact design
- Medical grade

Oxygen Concentrator



Reliable oxygen therapy

Reliable oxygen concentrators available in 5 L and 10 L configurations with single and dual flow options, designed for clinical and homecare settings with low-noise operation.

5 L / 10 L
Flow configurations

Single / dual
Flow outputs

ISO 13485
Manufactured under

KEY FEATURES

- 5 L & 10 L flow
- Low-noise operation
- ISO 80601-2-69 compliance
- Single or dual flow
- Clinical & homecare use
- Made in a Class 10K cleanroom

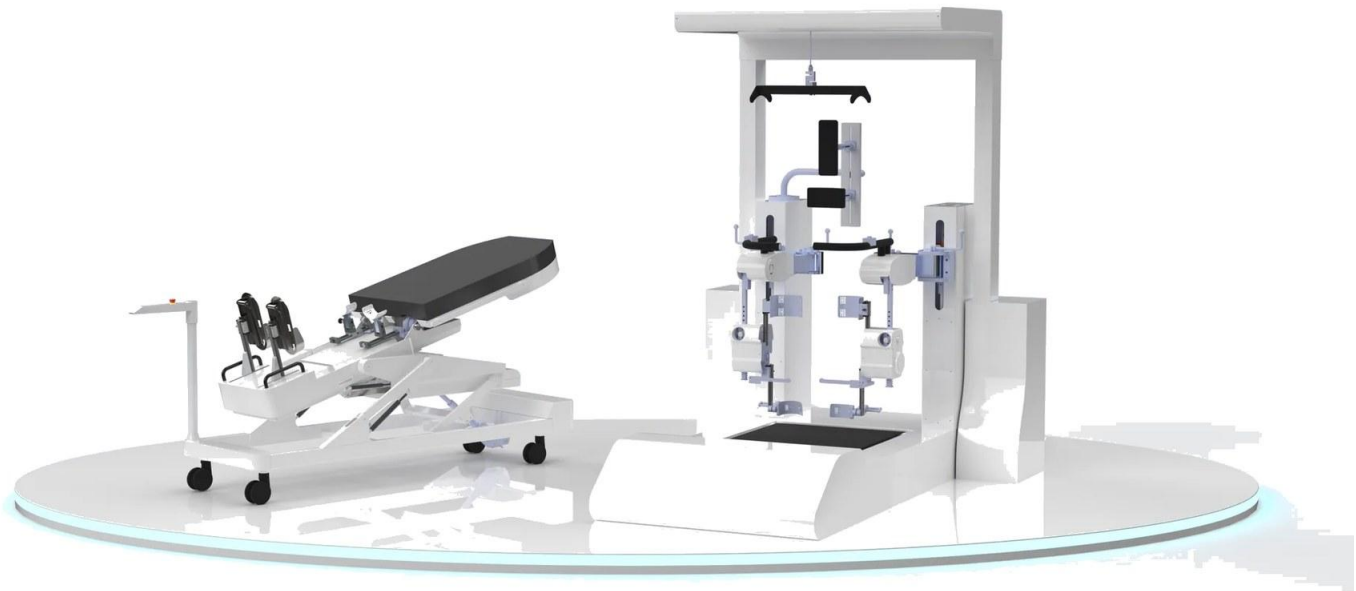
Walnut Technologies

05

Neuro-rehabilitation

Rehabilitation robotics and neuro-stimulation for stroke, traumatic brain injury, spinal cord injury and neurological conditions — with AI-assisted therapy and real-time monitoring.

AI-assisted therapy | Tele-rehab ready | Real-time monitoring



WalkLab Gait Training System



Flagship · lower-limb gait training system

A high-level gait training platform that improves gait outcomes, accelerates recovery and reduces cost of care through intensive, repetitive retraining of normal movement. Sensors track lower-limb kinematics and muscle activation to guide each step with the correct timing and force, with instant audio and visual feedback — for stroke, TBI, spinal cord injury, cerebral palsy, MS, Parkinson's and orthopedic conditions.

150 kg	1.8–4 km/h	<10 min
Max patient weight	Treadmill speed	Therapist setup time

KEY FEATURES

- Neural gait retraining
- Real-time audio & visual feedback
- VR multitask training
- Objective progress per session

Early rehabilitation & neuromodulation



Weebo

Early rehabilitation robot with FES

A motorised tilt table with an integrated leg-stepping mechanism. The table rises gradually toward vertical while the legs move in an alternating stepping pattern, maintaining venous return so verticalization can start earlier and be tolerated longer.

0–90°
Tilt range

45°
Leg stroke

20–39
Steps / min

- Progressive verticalization
- Alternating stepping motion
- Maintains venous return
- Prevents bed-rest complications



CES — Repose

Mobile cranial electrotherapy stimulator

Portable cranial electrotherapy stimulation device from the Walnut Medical neuro-modulation range. Also available in the range: Functional Electrical Stimulation (FES), rTMS and RMNS (right median nerve stimulation).

- Portable, mobile unit
- Multiple programmes & intensity levels
- Clear control panel
- Part of the neuromodulation range

Walnut Technologies

06

Fintech & payment devices

BIS-approved smart payment soundboxes with multi-language audio confirmation, secure integrations and a complete software ecosystem — in mass production since 2023.

BIS approved | PCI-DSS | NPCI compliant | EMV L1/L2



Soundbox hardware platform



PLATFORM SPECIFICATIONS

Types	Tabletop, pocket, DQR
Connectivity	LTE CAT 1 — 4G, 3G, 2G, Bluetooth, Wi-Fi
Battery	Rechargeable, up to 6000 mAh — lasts about 1 month per charge
Internal memory	4 MB to 256 MB
Audio output	2–5 W
SIM card	Nano / Micro
POS integration	API for POS integration
Add-on features	LCD with backlight, BT, Wi-Fi, FM, solar
Protection	Level 4 ESD protection, pin-hole reset
Power efficiency	PCB consumes 4 mA (vs 20–35 mA typical)

SOFTWARE & SERVICES

- **Finished hardware in your branding**
- **MQTT broker — low-latency announcements**
- **Terminal management system & OTA**
- **SIM connectivity partners**
- **Field deployment app (Android & iOS)**
- **500K+ units / month capacity**

Soundbox models



RTOS — All-in-one

Multiple display options to suit every merchant.

- 3.47″ or 2.8″ LCD, or segment display
- 2.4″ merchant display
- On-device QR generation
- Built-in keyboard



DQR — Digital QR

Compact digital-QR soundbox with a high-contrast LCD.

- Digital QR display
- Bright LCD for amount & confirmation
- Loud, clear audio
- Plug-and-play deployment



LDQR — Large display QR

Premium soundbox with a large touchscreen customer display.

- 10″ touchscreen
- 2.4″ merchant display
- Keyboard & custom QR
- NFC tap-and-pay



Payment soundbox

Smart payment speaker for every counter.

- 4G / 2G, NB-IoT
- Bluetooth & Wi-Fi
- Multi-language audio
- Long battery life

Walnut Technologies

07

Automotive electronics

Smart instrument clusters in 5-inch and 7-inch, Android and RTOS based, with 4G, Bluetooth and Wi-Fi — and Google Maps navigation through phone screen mirroring.

5″ & 7″ | Android · RTOS | 4G · BT · Wi-Fi | Phone mirroring



Smart Instrument Cluster



Digital clusters for 2-wheelers, tractors & farm equipment, LMV & HMV

Next-generation digital clusters with high-resolution displays, real-time vehicle data and customisable driver interfaces — built for durability and clarity. Real-time processing of speed, RPM, fuel, temperature and diagnostics, with CAN bus support.

KEY FEATURES

- High-resolution LCD/TFT, sunlight readable
- Low power consumption
- Customisable layouts & themes
- Advanced real-time processing
- Rugged, industrial-grade build
- Operates from −40 °C to +85 °C

IATF 16949 ready | **AEC-Q100** | **EMC tested** | **Vibration tested**

Choose your size: 5-inch or 7-inch



5″ **5-inch display**

Smart instrument cluster



7″ **7-inch display**

Smart instrument cluster

Clusters shown at relative scale.

CONFIGURATION OPTIONS

Display size	5-inch · 7-inch
Platform	RTOS · Android
Connectivity	4G · Bluetooth · Wi-Fi
Navigation	Phone screen mirroring — Google Maps on the cluster
Interface	Custom layouts, themes and brand integration
Vehicle data	Real-time speed, RPM, fuel, temperature, diagnostics
Vehicle bus	CAN bus support

Final specifications and combinations are confirmed per project; datasheets on request.

Google Maps on the cluster

Phone screen mirroring brings turn-by-turn Google Maps navigation from the rider's phone onto the cluster display — directions stay in the line of sight, alongside speed and vehicle data.



Turn-by-turn on the dash

Google Maps navigation from the rider's phone appears on the cluster screen.

Eyes stay on the road

Directions sit in the driver's line of sight — no phone on the handlebar or dashboard.

Works with your phone

Pair over the cluster's wireless connectivity and start navigating.

One screen for everything

Navigation, speed and vehicle data together on the cluster display.

Agri & off-highway electronics



Automatic tractor electronics suite: a complete suite of smart, IoT-enabled electronics for agricultural equipment, earth-moving equipment and tractors.

- Smart instrument clusters (5″ & 7″)
- Tractor automation electronics
- GPS tracker & gateway telematics
- Cloud dashboards, mobile apps, OTA

AUTOMOTIVE ENGINEERING CAPABILITIES

01 Display technology

LCD/TFT · anti-glare · sunlight readable

02 Real-time data

CAN bus · diagnostics · instant updates

03 Custom design

Custom layouts · brand integration · DFM

04 Quality

IATF 16949 ready · AEC-Q100 · EMC & vibration tested

Walnut Technologies

08

IoT solutions

GPS tracker & gateway, 4G smart lock and the cloud and mobile platform behind them — designed, built and supported in-house.

GPS tracking | Telemetry | Fleet management | Cloud & apps



GPS Tracker & Gateway



Asset tracking & telemetry

Reliable GPS tracking device and gateway for industrial and logistics applications — real-time tracking, telemetry reporting and fleet management with secure cloud connectivity.

KEY FEATURES

- Real-time GPS tracking
- Telemetry reporting
- Fleet management
- Secure cloud gateway
- Mobile app ready (iOS & Android)
- Encrypted data transmission

Fleet & logistics **Asset tracking** **Industrial** **Agriculture**

4G smart lock & IoT platform



4G Smart Lock

Connected smart lock for logistics, assets and infrastructure — remote access control, real-time monitoring and cloud management from anywhere.

- Real-time status monitoring
- 6 months standby capability
- Remote access control
- Industrial-grade IP67 design
- 4G · BLE · Wi-Fi connectivity
- Cloud integration

IOT ENGINEERING, END TO END

Product design

Hardware · RTOS firmware · BLE / Wi-Fi / LTE

Security

Encryption · secure element · tamper detection

Manufacturing

SMT & PCB · RF testing · firmware flashing

Cloud & mobile

Dashboards · mobile apps · OTA updates

AWS · MQTT · HTTPS · FOTA · JSON · RTOS · BLE · 4G LTE · NB-IoT

Engineering expertise

Medical electronics

Life-critical device design with full regulatory compliance.

Embedded systems & IoT

Firmware, RTOS and cloud-connected embedded solutions.

PCB design & fabrication

Multi-layer, HDI and flex PCBs for high-volume production.

Manufacturing & assembly

4 SMT lines, 500K+ units per month, 99.8% yield.

Quality & compliance

ISO 13485 quality management, Class 10K cleanroom.

Payment hardware security

PCI PTS, EMVCo and NPCI certified terminal design.

Software & firmware

Full-stack firmware, drivers and companion apps.

Supply chain & logistics

Global component sourcing and door-to-door delivery.

WHY MANUFACTURERS CHOOSE WALNUT

Mass manufacturing

500K+ units every month

DICE & chip bonding

Maximum cost savings

Lean manufacturing

Across the whole organisation

Cloud & firmware in-house

AWS, RTOS, MQTT, FOTA

Debt-free & bootstrapped

A stable long-term partner

Fast growth

CAGR above 100% (last 4 years)

Trusted by leading brands

HDFC BANK

SBI

paytm

BharatPe

Apollo HOSPITALS
TOUCHING LIVES

INDUSTRIES WE SERVE

Healthcare
Hospitals, clinics and medical device companies

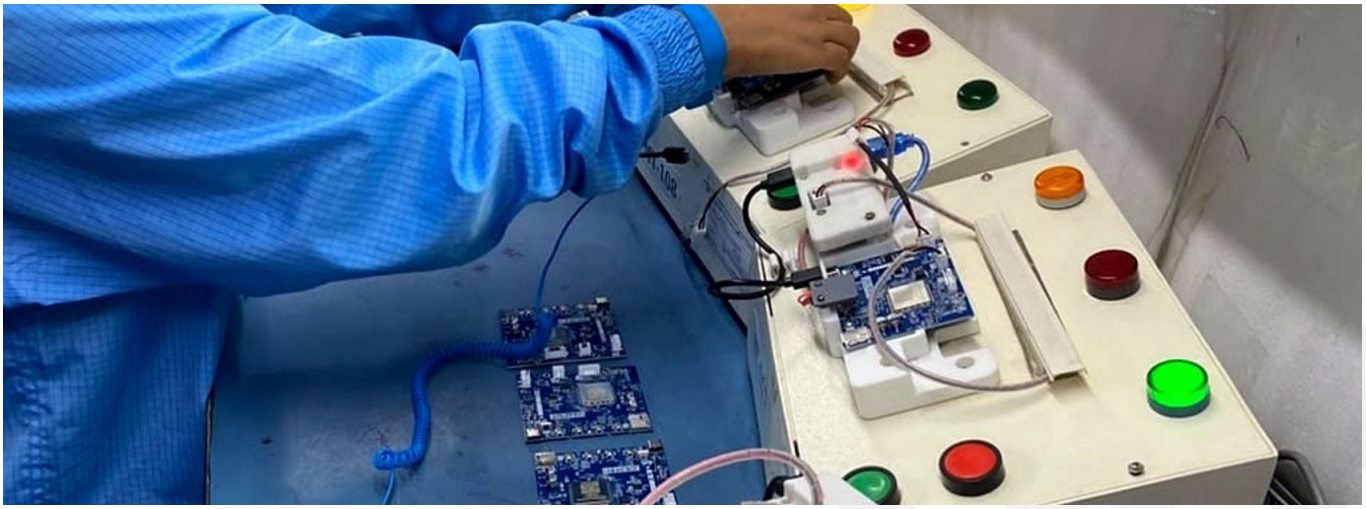
Defence & aerospace
Mission-critical electronics for defence

Banking & payments
POS terminals, soundboxes and payment devices

Pharmaceutical
Manufacturing equipment and monitoring systems

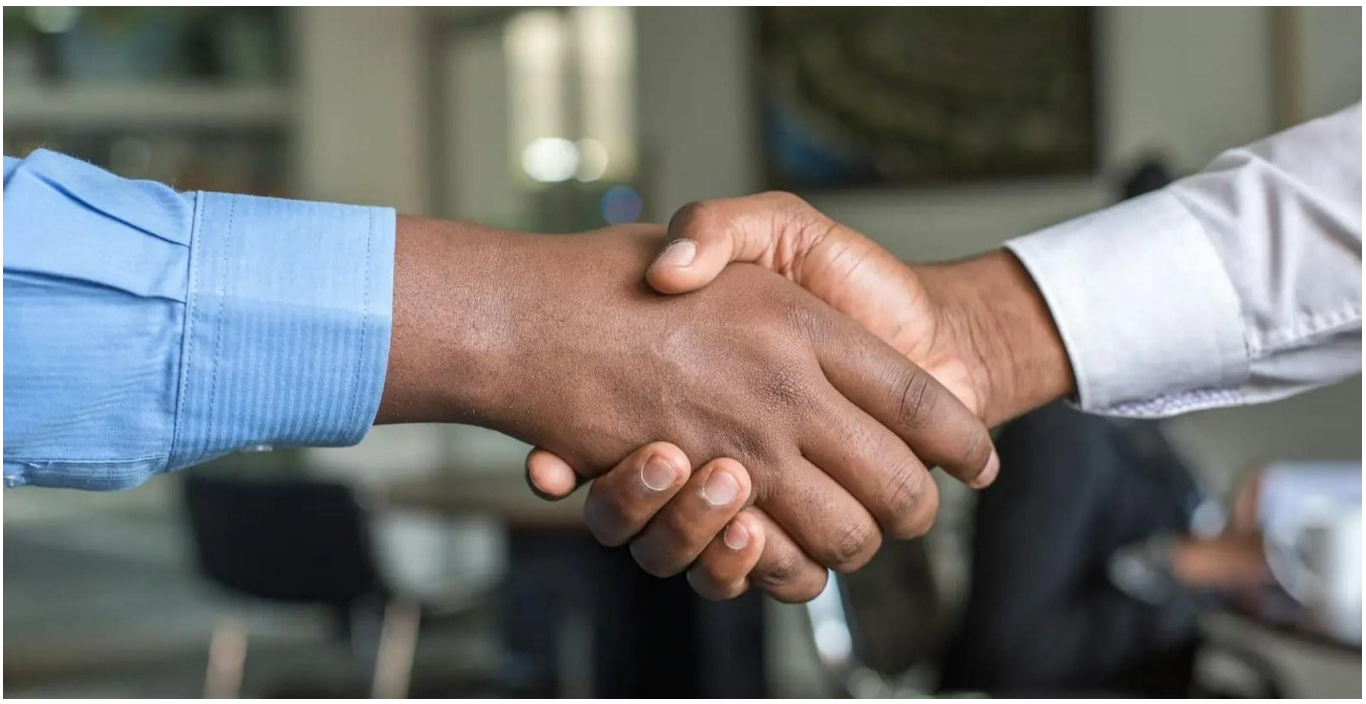
Energy & utilities
Smart metering, grid monitoring, industrial controls

Industrial & automotive
Industrial electronics, sensors and automotive



Let's build what's next

Engineering precision with user-centric design. We welcome the opportunity to collaborate with your team — from a first prototype to mass production.



Address

Plot No. 132, JLPL Industrial Park, Sector 82, Mohali, Punjab 160055, India

Upcoming facility

Dhanana (Upcoming in Haryana)

Phone

+91 77194 63719

Email

contact@walnutmedical.in

Website

walnutmedical.in

Notes

Walnut Technologies™

Intelligent Embedded Systems, Engineered for a Connected World

Medical · Neuro-rehabilitation · Fintech · Automotive · IoT

Walnut Technologies Pvt. Ltd.
Plot No. 132, JLPL Industrial Park, Sector 82, Mohali, Punjab 160055, India
Dhanana (Upcoming in Haryana)
+91 77194 63719 · contact@walnutmedical.in · walnutmedical.in